Engineering Mathematics - I · Unit_03 · Unit 3

DRAFT — UE26MA141A · Unit_03 · Unit 3 · Higher order PDE

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Higher order PDE

1 Higher order PDE

Study focus: Higher order PDE. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000118 · EV0000119

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In this session, we will discuss the solution of higher order linear partial differential equations with constant coefficients.

Key details

- In second chapter, we understood how to solve homogeneous and non-homogeneous higher order linear ordinary differential equity constant coefficients.
- Those methods can be extended to solve higher order linear partial differential equations with constant coefficients.
- That is aught times dou nz divided by dou x n plus.
- A1 times dou n z divided by dou x n minus 1 dot dou y and so on.
- And if you see on the right hand side we have f of x y.
- Power R of Z equals f of x y.
- Remember, from now onwards we use these notations.
- Small p is nothing but douz by dou x, which is same as dz, q same as douz by dou y.
- This is nothing but d dash up z.
- And this is r value.
- And this is S value.
- And similarly we have a T to be dou square Z divided by dou y square.
- This is nothing but D dash square of Z.
- We will now understand what is the general solution of this expression number two.
- Similar to the ordinary differential equations, the general solution of equal can be written as Z equals complementary function that is C dot F plus particular integral PI Where complementary function is nothing but the solution of.

- Homogeneous equation that is f of d comma d dash z equals zero, that is left hand side, and the particular integral is a solution satisfying.
- A non-homogeneous equity that is two and it doesn't contain any arbitrary constants.
- Before we understand how to compute complementary function, we need to know there is something called reducible factor and irreducible factor.
- Note that if f of d, d dash can be factorized into linear factor in the form AID plus VID dash plus c i, then this f of b, d dash is said to be reducible.

Example or application from the lecture

- Let's look at the general linear partial differential equation of order n in two variables.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000118 · EV0000119 · EV0000120 · +1 source units in pack.json

Core explanation · 05:00–09:59

Let's look at d square minus a d dash square.

Key details

- As, you can use a square minus b square formula.
- This exactly looks like.
- AID plus BID dash and of course C0 it is also acceptable.
- D minus 2d dash and d minus 3d dash up zero.
- Therefore we say that this f of d comma d dash up z 0 is reducible.
- You can clearly see that this is not reducible because you can't express this in the form of EID plus B ID dash plus CI.
- Even the third one is also.
- not reducible factor because you can't express this in terms of AID plus BID dash plus ci.
- D power three minus d square d dash plus d dash minus d dash square can be written as.
- D minus d dash times d square plus d dash square up z zero.
- It is easy to see that from the first two we can take d square common and you get d minus d dash.
- From third and fourth term you can take a d dash common, you will get d minus d dash.
- The first two are reducible factors or reducible forms and equation six and seven are not reducible or irreducible forms.
- So that's the next question.
- Category number one, we may have distinct rules.
- Second category, we may have repeated rules or factors.
- Of course we call it as existing factors and repeated factors.
- This form that is aid plus bid dash plus c i of z zero.
- of BIX minus a i y.
- So if C i is zero, obviously the first term becomes one.
- So as a first case, if C i is zero, this is how we write the solution.
- In a similar manner, if a i is zero, we write the solution in this form.

Example or application from the lecture

- If you look at the second example, this can be written as D minus D dash.
- And what about the third example?
- What about the fifth example?

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EV0000122 · EV0000123 · EV0000124 · +1 source units in pack.json

Core explanation · 10:00–14:59

In a similar manner, if Bi is 0, this is how we write the solution.

Key details

- That is e power minus c1x by a1.
- times phi 1 p1x minus a1y and so on.
- It depends on number of factors present in the problem.
- A I D plus B I D dash plus CI.
- It is easy to see that a1 is 1, B1 is minus 1.
- For the second factor, A2 is 1 and B2 is 1.
- Therefore, we can use equation 11.
- So based on this idea, so we first get z equals phi one.
- of if we multiply p1 by x we get minus x then minus y.
- We write it as some phi one star of minus x minus y.
- If we take minus common, then that is written as phi one of x plus y.
- And there's no change in the second term that is phi two of x minus y.
- And we can call this a general solution because if you look at the problem.
- The order of the PD is 2, and the number of arbitrary functions are also 2.
- So therefore we can call this as general solution.
- Plus 11 D dash square minus 6 d dash 3 of Z 0.
- So remember there is a shortcut method.
- So that shortcut method is applicable only when we have scenario like this.
- If you observe.
- The total degree of each term is the same.
- If you look at this power value, this is 3 and 2 plus 1 is 3, and 1 plus 2 is 3, and this is 3.
- So it means so we can divide this expression by.
- As D dash three.
- So this is same as divide through by d dash power 3 so the first number is d by d dash power three then second number term is going to be b by d dash square.
- Then the third term is 11 d by d dash and the last one is six so z equals zero.
- p by d dash equals m then that equation becomes m power three minus six m square plus eleven m minus six zero and you can see this is a cubic equation in m.

Example or application from the lecture

- Let us take up these four examples.

- If you look at d square minus d dash square of z zero, as we know from the example, this can be written as d plus d dash and d minus d dash of z zero.
- So let's look at the second example that is d power tree minus 6 d square dash 10 d dash.

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EV0000126 · EV0000127 · EV0000128 · +1 source units in pack.json

Core explanation · 15:00–19:59

For example, when we write d by d dash equals 1.

Key details

- we get d equal to d dash so it is same as d minus d dash so one factor is d minus d dash similarly when m is 2 we get d by d dash equals 2 so that is same as d minus 2 d dash so this is second factor.
- At last you can write d by d dash equals 3.
- And you can see all these three are reducible factors.
- We write the solution in this form.
- And very well when the first two terms are negative, we just introduce a star, then we can take minus common that is same as phi one of x plus y, phi two of two x plus y and phi three of the x plus y.
- If all terms in f of d gamma d dash have the same total degree, then we can set d by d dash equals m and then we find the factors of the equation using m.
- This shortcut method cannot be applicable for the third and fourth examples.
- In third and fourth, the total degree of each term is not same.
- So we cannot apply shortcut method.
- Then how do we solve?
- So the above mentioned note doesn't applicable to the following two problems because the total degree of each term is not same.
- Then how do you find out the factors?
- So that should be done by a normal inspection method.
- And the total degree is 2, this is 2, and third is 2, and this is 1, this is 1, this is 0.
- 7 d dash can be written as D six d dash plus 1d dash.
- There's no change in two d dash square.
- Again this 7d can be written as 6d plus d.
- Again this 4D dash is same as two d dash plus two d dash plus two of z right so if you observe there are nine terms.
- So this can be further written as if we take D common, you will get 3D plus e dash plus one, it means you're combining this, this, and this.
- And again, if you can take two d dash common.
- Z so from here you can get this again because three D plus D dash plus one.

Example or application from the lecture

- So I hope this note is clear after looking at this second example.
- For example, in this case, we write the first term same.

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EV0000130 · EV0000131 · EV0000132 · +1 source units in pack.json

Core explanation · 20:00–24:59

And it is easier to see that the order of the PD is 2 and there are two factors here after finding a1 b1 c 1 and a2 b2 c2 and substitutes for equation.

Key details

- we get this as a general solution e power minus x pi 3 phi 1 of x minus 3y plus e power minus 2x phi two of two x minus y.
- Here phi one and phi two are any arbitrary functions.
- That is 2d square minus d dash minus d dash square minus d plus d dash.
- Again you can observe.
- The total degree of each term is not same.
- We can split dd dash to be minus two plus one and there's no change in the other three terms.
- So this is what we get in the next step.
- If you look at odd places, you can take d common.
- If you look at e1 places, take minus t dash common.
- We get 2D plus D dash minus 1 of Z 0.
- For the first factor, a1 is 1, B1 is minus 1, C1 is 0.
- For the second factor, a2, b2, c2 are 2 1 minus 1 respectively.
- Because b1 is minus 1, we get a star that is same as phi one of x plus y.
- Then for the second because c is non-zero, we get a power x by 2 because.
- C is minus 1 minus of minus that is plus and a2 is 2, that's why x by 2, by 2 of x minus 2 y.
- when we have distinct factors or different factors.
- You have taken two for understanding theory part.
- It could be three, four, and so on.
- If BI0 we get this, if AI0 we get this.
- CI0 is a part of this.
- When we have a factor in this form, so this is how we write the resolution.
- When BI0, this is how we write the solution.
- we write the general solution in this form.

Example or application from the lecture

- Let's take another example.
- So this should be solved like the previous example.
- Suppose assume that let a factor be of multiplicity too.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000134 · EV0000135 · EV0000136 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Higher order PDE transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/007_Higher_order_PDE__d750329ea5ff8f74.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Higher order PDE transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/007_Higher_order_PDE__d750329ea5ff8f74.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Higher order PDE transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/007_Higher_order_PDE__d750329ea5ff8f74.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9995; only the JSON segmentation supplies the displayed timestamps.
Higher order PDE transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/007_Higher_order_PDE__d750329ea5ff8f74.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.

2. 1 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
3. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
4. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
5. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
6. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
7. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00038	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0055. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00039	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0055. Sidecar variants compare...
SRC00040	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0055. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00041	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0055. Sidecar variants compare...

C Revision and retrieval practice